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ALFIDO_TECH_CUSTOMER_...

ecommerce_customer_dat × | Alfido_tech_Customer_Beh × | Launcher

Code

No Headings

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[16]:

```
# 1. convert 'Purchase Date' into actual datetime objects
df ['Purchase Date'] = pd.to_datetime(df['Purchase Date'])

# 2. Set our reference date (1 day after latest purchase i
snapshot_date = df['Purchase Date'].max() + pd.Timedelta(d

# 3. Group by Customer ID to calculate Recency,Frequency,
rfm = df.groupby ('Customer ID') .agg({
     'Purchase Date': lambda x: (snapshot_date - x.max()).
     'Customer ID': 'count' ,
     'Total Purchase Amount': 'sum'
})
# Rename columns for clarity
rfm.rename(columns={'Purchase Date': 'Recency' , 'Customer

# 4. Break scores down into quintiles ( 1 to 5 )
rfm['R_Score'] = pd.qcut(rfm['Recency'], 5, labels=[5, 4,
rfm['F_Score'] = pd.qcut(rfm['Frequency'].rank (method='fi
rfm['M_Score'] = pd.qcut(rfm['Monetary'], 5, labels=[1, 2,

#combine individual scores into an RFM cell string
rfm['RFM_cell'] = rfm['R_Score'].astype(str) + rfm ['F_Sco


# 5. Define customer behavior labels
def assign_segments (row):
 if row['RFM_cell'] in ['555' , '554' , '545' , '544']:
  return 'Champions (Best Customers)'
 elif row['R_Score'] == 5:
  return 'New Customers'
 elif row['R_Score'] in [1, 2] and row['F_Score'] in [4, 5
```

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